cvf.py - C:/Users/Admin/cvf.py (3.13.14)

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```
import numpy as np
from scipy.linalg import hadamard

W = hadamard(4)                    # 4 orthogonal Walsh codes (rows), +/-1
bits1 = np.array([1, -1, 1, 1])   # user 1 data (+/-1)
bits2 = np.array([-1, -1, 1, -1]) # user 2 data

def spread(bits, code):            # each bit multiplied by the whole code
    return np.concatenate([b * code for b in bits])

tx = spread(bits1, W[1]) + spread(bits2, W[2])  # channel adds signals

def despread(rx, code):
    chips = rx.reshape(-1, len(code))
    return np.sign(chips @ code / len(code))

print("User1 recovered:", despread(tx, W[1]), "sent:", bits1)
print("User2 recovered:", despread(tx, W[2]), "sent:", bits2)
print("Wrong code W3 correlation:", (tx.reshape(-1, 4) @ W[3]) / 4)  # all zeros
```

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```
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit
 (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>
========== RESTART: C:/Users/Admin/cvf.py ==========
C:\Users\Admin\AppData\Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Pyth
on.3.13_qbz5n2kfra8p0\pythonw.exe
>>
========== RESTART: C:/Users/Admin/cvf.py ==========
User1 recovered: [ 1. -1.  1.  1.] sent: [ 1 -1  1  1]
User2 recovered: [-1. -1.  1. -1.] sent: [-1 -1  1 -1]
Wrong code W3 correlation: [0. 0. 0. 0.]
>>
========== RESTART: C:/Users/Admin/cvf.py ==========
User1 recovered: [ 1. -1.  1.  1.] sent: [ 1 -1  1  1]
User2 recovered: [-1. -1.  1. -1.] sent: [-1 -1  1 -1]
Wrong code W3 correlation: [0. 0. 0. 0.]
>>
```